Automation of Patient Specific Breast Cancer Image Segmentation Using 2D U-net MONAI

K S Suraksha Jain [1] | Guide: Prof. Kavitha Arunachalam [2]

[1] Department of Applied Mechanics & Biomedical Engineering | [2] Department of Engineering Design | Indian Institute Of Technology Madras

INTRODUCTION

Breast cancer is one of the most prevalent cancers worldwide, making accurate tumor localization essential for diagnosis, treatment planning, and therapeutic interventions. Magnetic Resonance Imaging (MRI) provides excellent soft-tissue contrast and detailed anatomical information, enabling effective visualization of breast tumors. However, manual tumor segmentation is ***labor-intensive, time-consuming, and prone to inter-observer variability***, which can affect consistency and clinical efficiency.

To address these challenges, this study proposes an automated patient-specific breast tumor segmentation framework using a two-dimensional ***2D U-Net architecture*** implemented in the ***MONAI framework***. The proposed approach aims to ***improve segmentation accuracy, reduce clinical workload***, and ***support future image-guided breast cancer therapy applications.***

OBJECTIVE

- Automated Breast Tumor Segmentation
- MRI Volume Resampling & Padding
- Normalization Technique Comparison
- Segmentation Performance Evaluation
- False Positive Reduction in Post Processing

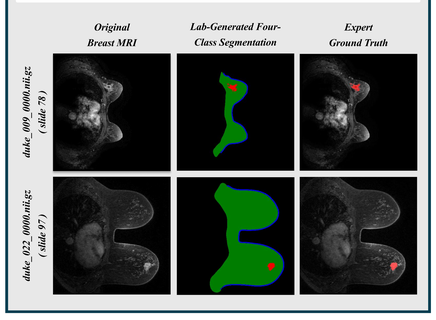
DATASET

DUKE Brest Cancer MRI Dataset

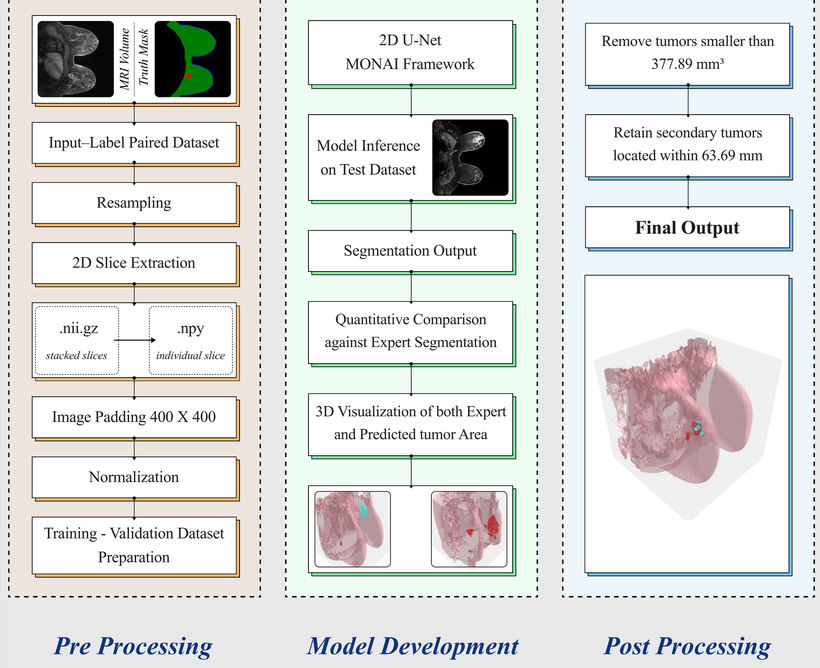
- 30 MRI volumes for training
- 10 MRI volumes for testing

5114 Total → ***4110 Train Data*** ***1004 Validation Data***

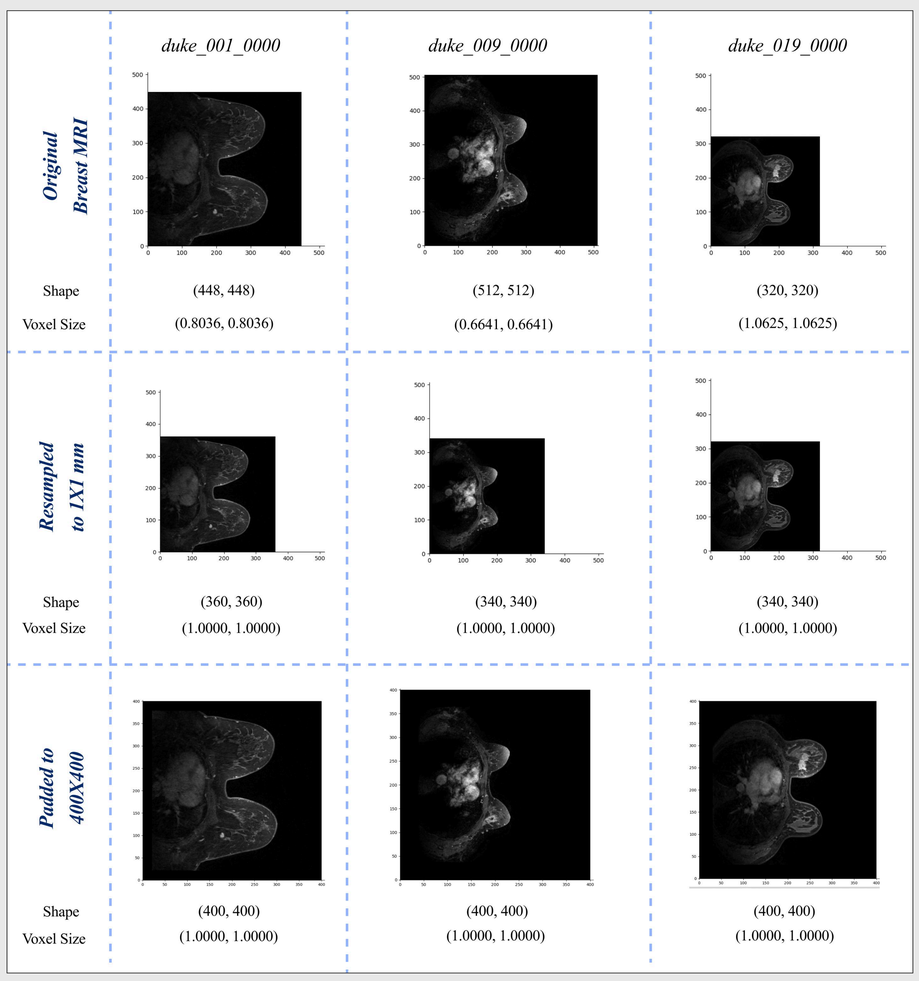
Class Label	Assigned Label ID (Color)
Background	0 (Black)
Tumor	1(Red)
Glandular tissues	2 (Green)
Skin	3 (Blue)

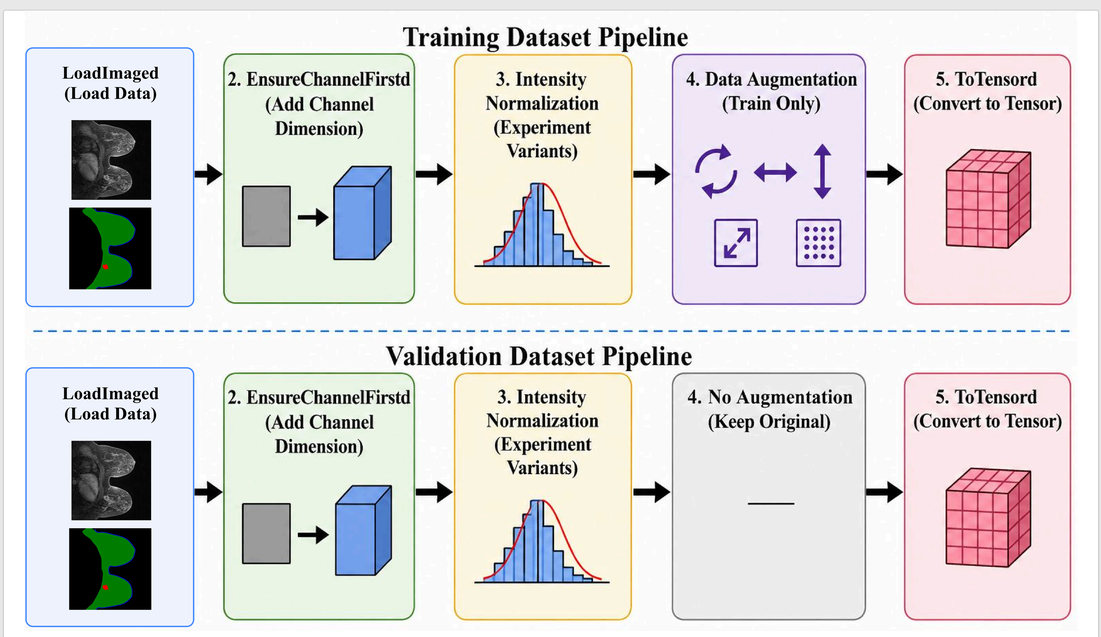


WORKFLOW

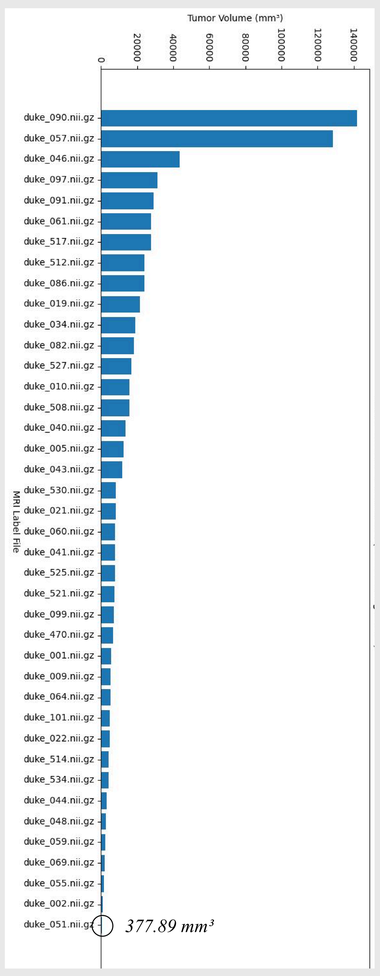


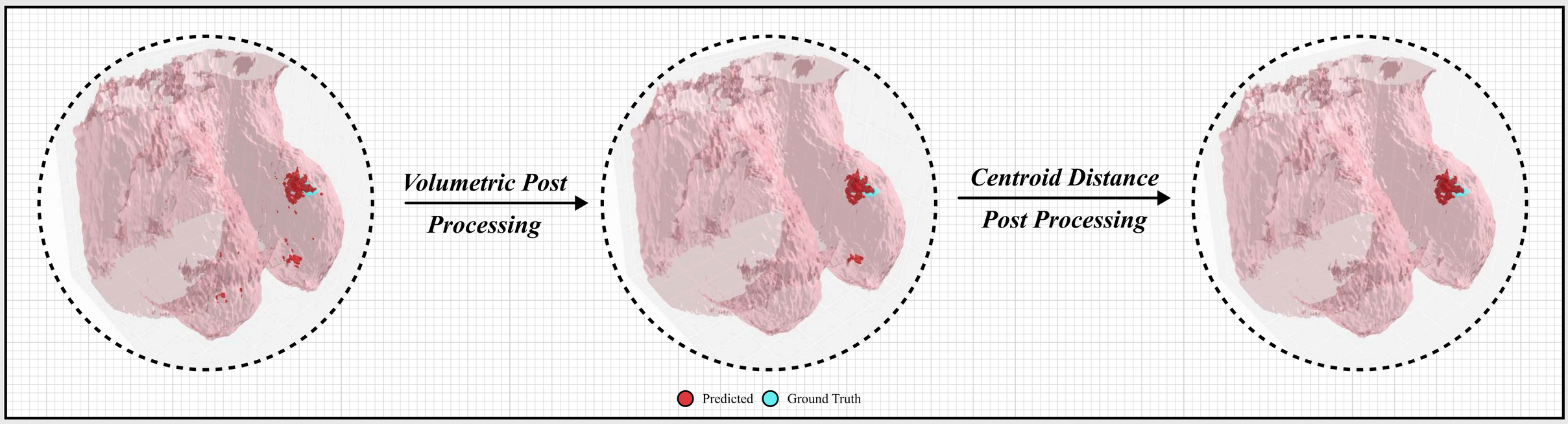
RESULTS





Method	Best Validation Dice	TestDice	TestIoU	TestPrecision	TestRecall
No Normalization	0.4751	0.3242	0.1987	0.1987	1
Min-Max	0.4744	0.5702	0.416	0.4421	0.945
Min-Max (Non-zero)	0.5421	0.4973	0.3433	0.3433	1
Z-score	0.6906	0.5861	0.441	0.4724	0.914





CONCLUSION

This work presents an automated patient-specific breast tumor segmentation framework using a MONAI-based 2D U-Net architecture. A comprehensive pipeline incorporating resampling, slice extraction, padding, intensity normalization, and two-stage post-processing was developed for breast MRI analysis. Experimental results demonstrated that appropriate normalization significantly influences segmentation performance, while post-processing effectively reduced false-positive detections. The proposed framework provides a computationally efficient and reliable solution for breast tumor localization and can serve as a foundation for future image-guided breast cancer therapeutic planning. Furthermore, the developed segmentation pipeline demonstrates the potential of deep learning–based approaches to enhance clinical decision-making and support personalized breast cancer treatment strategies.

FUTURE WORK

- Optimize the number of feature channels to improve feature extraction and segmentation performance.
- Investigate different network depths to balance segmentation accuracy and computational cost.
- Evaluate the effect of varying residual convolutional units on model performance and training stability.
- Incorporate dropout-based regularization to reduce overfitting and improve model generalization.
- Assess alternative activation functions to enhance feature learning and network convergence.

REFERENCES

- L. Garrucho, K. Kushibar, C. A. Reidel, S. Joshi, R. Osuala and A. Tsirikoglou, "A large-scale multicenter breast cancer DCE-MRI benchmark dataset with expert segmentations," Scientific Data, p. 453, 2025.
- M. J. Cardoso, W. Li, R. Brown, N. Ma, E. Kerfoot and Y. Wang, "MONAI: An open-source framework for deep learning in healthcare," 2022.M. J. Cardoso, W. Li, R. Brown, N. Ma, E. Kerfoot and Y. Wang, "MONAI: An open-source framework for deep learning in healthcare," 2022.

ACKNOWLEDGEMENT

Dedicated to the Almighty, my Supervisor Prof. Kavitha Arunachalam, my parents, my lab mates, my friends